DTC P081D:29 [START STOP UNIT]

id0902p6022900

Description	Neutral switch error signal received
Detection condition	• Start stop unit receives the neutral switch signal error signal from the PCM for **1 min or more** with the ignition switched ON (engine off or on).
Fail-safe	• Perform the control with neutral switch signal turned off.
Possible cause	• DTCs are stored in the PCM. • Neutral switch malfunction • PCM malfunction • Start stop unit malfunction
System wiring diagram	Not applicable

Diagnostic Procedure

Step	Inspection		Action
1	**VERIFY PCM DTCs** • Retrieve the PCM DTCs using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Are any DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC TABLE [SKYACTIV-D 1.5].)
		No	Go to the next step.
2	**INSPECT NEUTRAL SWITCH** • Inspect the neutral switch. (See NEUTRAL SWITCH INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See NEUTRAL SWITCH INSPECTION [SKYACTIV-D 1.5].) • Is the neutral switch normal?	Yes	Go to the next step.
		No	Replace the neutral switch, then go to Step 4. (See NEUTRAL SWITCH REMOVAL/INSTALLATION [F65M-R].) (See NEUTRAL SWITCH REMOVAL/INSTALLATION [F66M-R].)
3	**VERIFY IF MALFUNCTIONING LOCATION IS PCM DEPENDING ON REPEATABILITY** • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Switch the ignition ON (engine off or on) and wait for **1 min or more**. • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Replace the PCM, then go to the next step. (See PCM REMOVAL/INSTALLATION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
		No	Go to Step 5.
4	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Clear the DTC for the start stop unit using the M-MDS. (See CLEARING DTC [START STOP UNIT].) • Switch the ignition ON (engine off or on) and wait for **1 min or more**. • Retrieve the start stop unit DTCs using the M-MDS. (See DTC INSPECTION [START STOP UNIT].) • Is the same DTC displayed?	Yes	Repeat the inspection from Step 1. • If the malfunction recurs, replace the start stop unit. (See START STOP UNIT REMOVAL/INSTALLATION.) Go to the next step.
		No	Go to the next step.
5	**VERIFY IF OTHER DTCs DISPLAYED** • Are any other DTCs displayed?	Yes	Repair or replace the malfunctioning part according to the applicable DTC troubleshooting. (See DTC TABLE [START STOP UNIT].)
		No	DTC troubleshooting completed.